SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	03-08-2026	**Duration:**	60 minutes

Code of Conduct

I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Malli chandana

Annexure A – Scenario-Based Requirement Analysis

Software Requirement Specification (SRS) for Eco-Tourism Management and Sustainable Travel Recommendation System

1 – INTRODUCTION

1.1 Introduction

The **Eco-Tourism Management and Sustainable Travel Recommendation System** is a software solution that promotes responsible tourism while protecting the environment and supporting local communities. Many existing tourism platforms mainly focus on popular destinations and provide limited information about eco-friendly travel options. Tourists often face difficulties in finding sustainable destinations, green accommodations, and personalized travel recommendations. The proposed system integrates destination information, booking facilities, travel recommendations, and user feedback into a single platform. Using technologies such as Artificial Intelligence (AI), cloud computing, and GPS, the system helps users plan environmentally friendly trips while encouraging sustainable tourism. It also supports tourism authorities in managing information and improving visitor experiences.

Key Points

- Promotes sustainable tourism.
- Supports eco-friendly travel planning.
- AI-based recommendations.
- Encourages environmental conservation.
- Integrates booking services.
- Improves tourist experience.
- Supports local communities.

1.2 Purpose of the SRS

The purpose of this **Software Requirement Specification (SRS)** is to define the requirements of the **Eco-Tourism Management and Sustainable Travel Recommendation System**. It serves as a communication document between stakeholders, developers, testers, and project managers. The SRS describes the system objectives, functional requirements, non-functional requirements, and technical specifications needed for software development. It also acts as a reference throughout the Software Development Life Cycle (SDLC), ensuring that all stakeholders understand the project clearly. A well-defined SRS improves software quality and supports future maintenance.

Key Points

- Defines system requirements.
- Guides software development.
- Supports communication.
- Helps testing and maintenance.
- Reduces misunderstandings.
- Improves software quality.
- Ensures stakeholder agreement.

1.3 Problem Description

Sustainable tourism has become increasingly important, but many travel platforms provide limited support for eco-friendly tourism. Tourists often need to visit multiple websites to gather information about destinations, accommodation, transportation, and bookings. This process is time-consuming and may not provide reliable recommendations. Tourism authorities also face challenges in promoting eco-tourism and managing destination information effectively. The proposed system solves these issues by providing a centralized platform that offers personalized recommendations, secure booking, and sustainable travel guidance.

Key Points

- Lack of integrated tourism platform.
- Limited eco-tourism information.
- Multiple websites required.
- Difficult travel planning.
- Low awareness of sustainable travel.
- Need for centralized solution.
- Supports responsible tourism.

1.4 Scope of the System

The proposed system provides a digital platform for managing eco-tourism activities. Users can register, search destinations, receive AI-based recommendations, book accommodations, and provide feedback. Administrators can manage users, destinations, bookings, and reports through a centralized dashboard. The system supports secure online transactions and can be accessed through both web and mobile devices. Future enhancements may include technologies such as Augmented Reality (AR), IoT, and predictive analytics to further improve the tourism experience.

Key Points

- Destination management.
- AI recommendations.
- Online booking.
- User feedback.
- Admin dashboard.
- Web and mobile support.
- Future technology integration.

1.5 Definitions, Acronyms, and Abbreviations

The following terms are used throughout this Software Requirement Specification document to ensure consistency and a clear understanding of the proposed system.

Term	Meaning
SRS	Software Requirement Specification
AI	Artificial Intelligence
SDLC	Software Development Life Cycle
UI	User Interface
GPS	Global Positioning System
DBMS	Database Management System
API	Application Programming Interface
Admin	System Administrator

1.6 References

This SRS is prepared using software engineering concepts, Agile development practices, IEEE SRS guidelines, and research related to eco-tourism management. These references provide the foundation for designing a secure, scalable, and user-friendly software solution.

Key Points

- IEEE SRS Guidelines.
- Software Engineering.
- Agile Scrum.
- Eco-tourism research.
- AI concepts.
- Sustainable tourism.
- Web application practices.

1.7 Document Overview

This document explains the complete software requirements of the proposed system. It includes scenario analysis, functional and non-functional requirements, use cases, data requirements, system features, assumptions, constraints, and requirement validation. The document provides a structured approach for developing and maintaining the software successfully.

Key Points

- Introduction.
- Scenario analysis.
- Functional requirements.
- Non-functional requirements.
- Use case modelling.
- Requirement validation.
- Future enhancements.

2 – STUDY AND ANALYZE THE SCENARIO

2.1 Business Domain

The **Eco-Tourism Management and Sustainable Travel Recommendation System** belongs to the **Tourism and Hospitality** domain. It focuses on promoting eco-friendly tourism by helping travelers discover sustainable destinations, green accommodations, and environmentally responsible travel options. The system also supports tourism authorities and local communities by encouraging responsible tourism and improving destination management through digital technology.

Key Points

- Tourism and Hospitality domain.
- Promotes eco-tourism.
- Supports sustainable travel.
- Helps local communities.
- Encourages environmental conservation.
- Improves tourism management.
- Digital tourism platform.

2.2 Stakeholders

The proposed system involves several stakeholders who contribute to its successful operation. The primary stakeholders are tourists, who use the platform to search destinations, receive travel recommendations, and book trips. Tourism authorities and administrators manage destination information, bookings, and reports. Local businesses such as hotels and transport providers offer eco-friendly services, while software developers maintain and improve the platform. Environmental organizations also benefit by promoting sustainable tourism practices.

Key Points

- Tourists.
- Administrators.
- Tourism authorities.
- Local businesses.
- Hotels and transport providers.
- Software developers.
- Environmental organizations.

2.3 Existing Challenges

Current tourism platforms mainly focus on commercial tourism and often provide limited information about eco-friendly destinations. Users need to search multiple websites for bookings, accommodation, and travel information. Personalized recommendations are limited, and awareness of sustainable tourism remains low. These issues reduce user convenience and make responsible travel planning more difficult.

Key Points

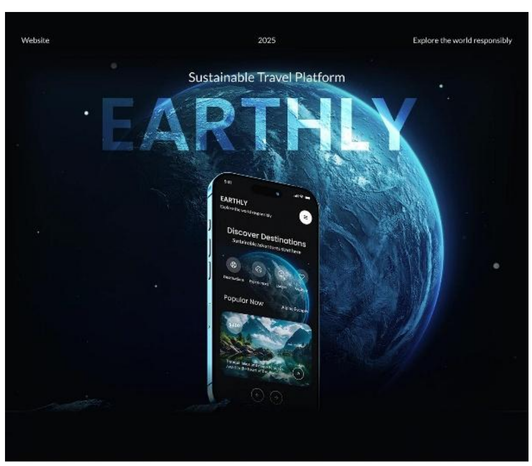
- Scattered travel information.
- Limited eco-tourism support.
- Multiple websites required.
- Lack of personalized recommendations.
- Low awareness of sustainable travel.
- Time-consuming planning.
- Poor integration.

2.4 System Objectives

The main objective of the proposed system is to provide a centralized platform for planning sustainable trips. It helps users discover eco-friendly destinations, receive AI-based travel recommendations, book accommodations, and access reliable travel information. The system also supports tourism authorities by simplifying destination management and promoting environmental conservation.

Key Points

- Promote sustainable tourism.
- AI-based recommendations.
- Easy travel planning.
- Secure online booking.
- Improve user experience.
- Support tourism authorities.
- Encourage environmental conservation.



3 – IDENTIFY FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

3.1 Functional Requirements

Functional requirements describe the core functions that the **Eco-Tourism Management and Sustainable Travel Recommendation System** must perform. These requirements ensure that users can access the platform, search for eco-friendly destinations, receive travel recommendations, book accommodations, and provide feedback. Administrators can manage users, destinations, bookings, and reports. These functions improve user experience while promoting sustainable tourism.

Key Points

- User registration and login.
- Search eco-tourism destinations.
- AI-based travel recommendations.
- Online booking and payment.
- User profile management.
- Ratings and reviews.
- Admin dashboard and report generation.

Functional Requirements

- **FR1:** User Registration.
- **FR2:** User Login and Authentication.
- **FR3:** Manage User Profile.
- **FR4:** Search Eco-Tourism Destinations.
- **FR5:** AI-Based Travel Recommendation.
- **FR6:** Book Hotels and Travel Packages.
- **FR7:** Secure Online Payment.
- **FR8:** Submit Ratings and Reviews.
- **FR9:** View Booking History.
- **FR10:** Administrator Manages Users and Destinations.
- **FR11:** Generate Reports.

- **FR12:** Send Notifications and Updates.

3.2 Non-Functional Requirements

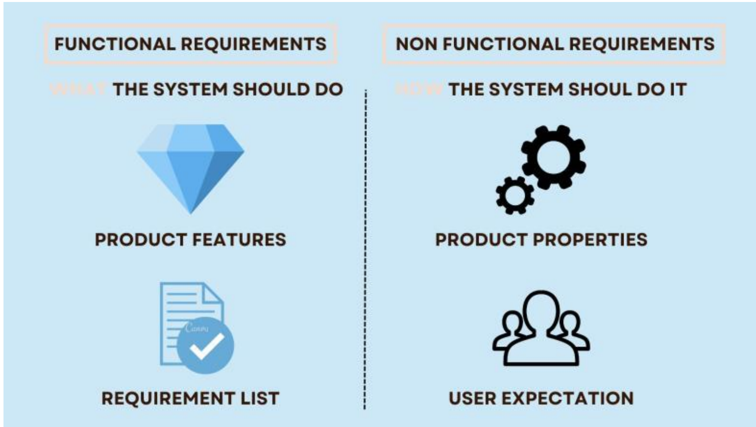
Non-functional requirements define the quality attributes of the system. The platform should provide high performance, security, reliability, usability, and scalability while ensuring that user information is protected. It should be easy to use, respond quickly, and support future enhancements without affecting system performance.

Key Points

- High performance.
- Secure user authentication.
- Reliable system operation.
- User-friendly interface.
- Scalable architecture.
- Easy maintenance.
- High availability.

Non-Functional Requirements

- **Performance:** Pages should load quickly and support multiple users.
- **Security:** Encrypt passwords and secure online payments.
- **Reliability:** Ensure minimum downtime and regular backups.
- **Usability:** Provide a simple and responsive interface.
- **Scalability:** Support future users and additional destinations.
- **Maintainability:** Easy updates and bug fixes.
- **Availability:** System should be available 24×7.



4 – SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Use Case Descriptions

A use case describes how different users interact with the **Eco-Tourism Management and Sustainable Travel Recommendation System** to perform specific tasks. The main actor is the tourist, who uses the system to register, search destinations, receive travel recommendations, book trips, and provide feedback. The administrator manages users, destinations, bookings, and reports. These use cases help developers understand the system's functionality and user interactions before implementation.

Key Points

- Defines user interactions.
- Shows system functionality.
- Helps software design.
- Supports requirement analysis.
- Improves system understanding.
- Guides developers.
- Simplifies testing.

Major Use Cases

Actor	Use Cases
Tourist	Register, Login, Search Destinations, View Recommendations, Book Trip, Make Payment, Give Feedback

Actor	Use Cases
Administrator	Manage Users, Manage Destinations, Manage Bookings, Generate Reports, Update Information

4.2 Data Requirements

The system stores and manages data related to users, eco-tourism destinations, bookings, payments, and feedback. Accurate and secure data management is essential for providing personalized recommendations and maintaining system reliability. The database should support efficient storage, retrieval, and updating of information.

Key Points

- User information.
- Destination details.
- Booking records.
- Payment details.
- Feedback and reviews.
- Administrator data.
- Report information.

4.3 External Interface Requirements

The system provides a simple and user-friendly interface that can be accessed through web browsers and mobile devices. Users interact with the system using graphical interfaces, while administrators manage the platform through a secure dashboard. The system also connects with payment gateways, GPS services, and AI recommendation modules to improve user experience.

Key Points

- Web interface.
- Mobile interface.
- Admin dashboard.
- Secure payment gateway.
- GPS integration.

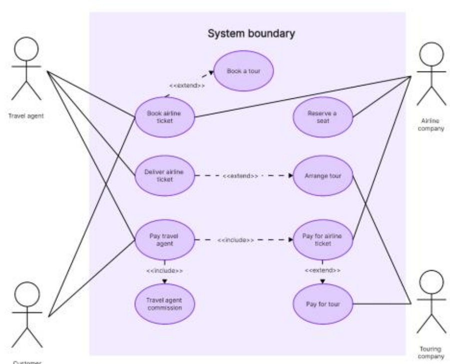
- AI recommendation service.
- Responsive design.

4.4 System Features

The proposed system includes several features that simplify eco-tourism management and sustainable travel planning. Users can search eco-friendly destinations, receive AI-based travel recommendations, book accommodations, make secure payments, and submit reviews. Administrators can manage destinations, monitor bookings, and generate reports. These features improve convenience, promote responsible tourism, and support environmental conservation.

Key Points

- User registration and login.
- AI travel recommendations.
- Eco-tourism destination search.
- Secure online booking.
- Online payment.
- Ratings and reviews.
- Admin management and reports.



5 – IDENTIFY ASSUMPTIONS AND CONSTRAINTS

5.1 Assumptions

The development of the **Eco-Tourism Management and Sustainable Travel Recommendation System** is based on several assumptions. It is assumed that users have internet access and basic knowledge of using web or mobile applications. Tourism authorities and administrators will regularly update destination information, while hotels and travel partners will provide accurate booking details. The system also assumes that users will provide genuine feedback and that third-party services such as payment gateways and GPS APIs will function correctly. These assumptions help ensure the smooth operation of the proposed system.

Key Points

- Users have internet access.
- Users possess basic digital skills.
- Destination information is regularly updated.
- Payment gateway services are available.
- GPS services function properly.
- Users provide genuine feedback.
- Administrators maintain the system.

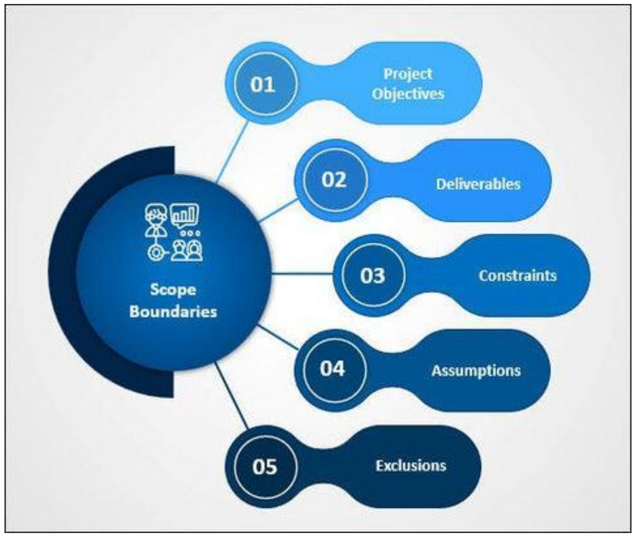
5.2 Constraints

The proposed system may face several constraints during development and implementation. Technical constraints include internet dependency and integration with third-party APIs. Economic constraints involve development and maintenance costs. Legal constraints require compliance with data privacy and online payment regulations. Time constraints may affect feature implementation within the project schedule. Operational constraints include regular content updates and system maintenance. Addressing these limitations is important for successful project delivery.

Key Points

- Internet dependency.
- Third-party API integration.
- Development cost.
- Data privacy regulations.

- Limited project timeline.
- Regular maintenance required.
- Resource availability.



6 – VALIDATE REQUIREMENT COMPLETENESS AND CONSISTENCY

6.1 Requirement Validation

Requirement validation ensures that all stakeholder needs are correctly identified and included in the Software Requirement Specification (SRS). The proposed **Eco-Tourism Management and Sustainable Travel Recommendation System** has been reviewed to confirm that the functional and non-functional requirements support the project objectives. The requirements are clear, realistic, and aligned with the needs of tourists, administrators, tourism authorities, and local businesses. Validation also helps reduce development errors and ensures that the system meets user expectations.

Key Points

- Requirements match stakeholder needs.
- Supports project objectives.
- Functional requirements verified.
- Non-functional requirements verified.
- Improves software quality.

- Reduces development risks.
- Ensures user satisfaction.

6.2 Completeness and Consistency Check

The proposed SRS has been checked to ensure that all important system functions are included and that there are no conflicting requirements. Features such as user registration, destination search, AI-based recommendations, online booking, secure payment, and administrator management are clearly defined. The requirements are organized logically and remain consistent throughout the document. This helps developers understand the system and implement it without confusion.

Key Points

- All major features included.
- Requirements are complete.
- No conflicting requirements.
- Consistent documentation.
- Easy to understand.
- Supports system development.
- Reduces implementation errors.

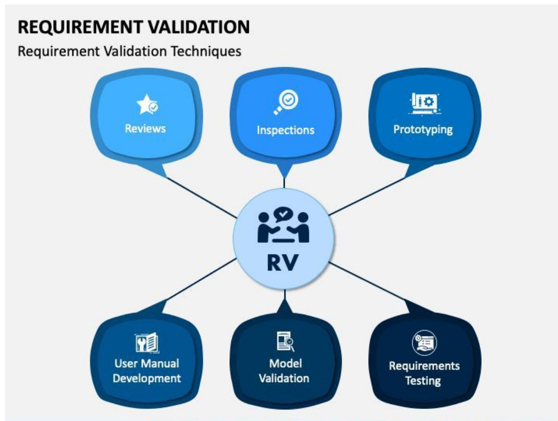
6.3 Suggested Improvements

Although the proposed system covers the essential requirements, future versions can include additional features to improve user experience. These enhancements may include multilingual support, Augmented Reality (AR) navigation, carbon footprint tracking, weather forecasting, social media integration, and advanced AI-based travel recommendations. Regular feedback from users and stakeholders can also help improve the system over time and ensure it remains effective for sustainable tourism.

Key Points

- Add multilingual support.
- Integrate AR navigation.
- Carbon footprint tracking.
- Weather updates.

- Social media integration.
- Advanced AI recommendations.
- Continuous user feedback.



Conclusion

The **Eco-Tourism Management and Sustainable Travel Recommendation System** provides a complete software solution for promoting sustainable tourism and responsible travel. The Software Requirement Specification (SRS) clearly defines the system objectives, functional requirements, non-functional requirements, actors, use cases, data requirements, assumptions, constraints, and validation process. The proposed system simplifies eco-friendly travel planning, improves tourism management, and supports environmental conservation through modern technologies. By following this SRS, developers can build a reliable, secure, scalable, and user-friendly application that meets stakeholder requirements and supports future enhancements.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5

Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25